



Thathsara Rajapaksha

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Profile

First Class Honours Data Science graduate with a track record of delivering production-ready analytics solutions. I specialize in building ML systems that scale (Azure, GCP), creating dashboards that inform decisions (Power BI, Tableau), and translating complex financial data into clear insights.

Skills

PROGRAMMING & STATISTICAL ANALYSIS: Python (Pandas, Numpy), R, SQL, KQL, Java, MATLAB, Statistical Modeling, Hypothesis Testing, A/B Testing

DATA VISUALIZATION & BUSINESS INTELLIGENCE: Power BI, Direct Lake, Tableau, Looker, Matplotlib, Seaborn, Plotly, Excel (Advanced), DAX, Dashboard Development, Stakeholder Reporting

MACHINE LEARNING & AI: TensorFlow, PyTorch, Keras, Scikit-learn, Deep Learning (CNNs, RNNs, Transformers), Transfer Learning, AutoML (Optuna), Uplift Modeling, Explainable AI (SHAP), Model Optimization (Quantization, Distillation, ONNX)

GENERATIVE AI & LLMs: OpenAI GPT, Anthropic Claude, Google Gemini, Meta Llama, AWS Bedrock, Hugging Face (BERT, DistilBERT, RoBERTa), RAG Architecture, Vector Databases (ChromaDB, Cosmos DB), LangChain, LangGraph, CrewAI, Workflow Automation, Agent/Copilot Design Patterns, Prompt Engineering, Agentic AI, Model Fine-tuning (LoRA)

CLOUD & MLOPS: Azure (ML Studio, Databricks, Fabric (OneLake, Eventhouse), Functions, Cosmos DB, Blob Storage, Data Factory), GCP (Vertex AI, BigQuery), AWS (SageMaker, Lambda, S3), Oracle, Supabase, Docker, Kubernetes, MLflow, CI/CD, Model Registry

DATA ENGINEERING: Apache Spark, PySpark, Airflow, Kafka, Hadoop, Snowflake, MS SQL Server, PostgreSQL, Delta Lake, Azure Event Hubs, ETL Pipelines, Real-Time Streaming, Polars, Data Annotation & Labeling, Data Quality Validation & QA Testing

FULL STACK & API: Next.js, React, TypeScript, Node.js, Tailwind CSS, Zustand, Framer Motion, FastAPI, Flask, Gradio, Streamlit, REST APIs, Microservices Architecture, HTML, CSS, JavaScript

Experience

AI/ML Engineer

Feb 2025 - Feb 2026

Codebell PVT LTD, Dikwella Rd, Beliatta, Sri Lanka

- Developed end-to-end ML pipelines for production projects across computer vision, NLP, and traditional ML domains, achieving around 80% model accuracy & delivering measurable business value through cross-functional collaboration
- Optimized models for production deployment using quantization (approx. 40% size reduction), distillation, caching strategies, and implemented fairness-aware ML with bias mitigation.
- Delivered production-ready models with comprehensive documentation, collaborating with product and business teams to ensure models met stakeholder requirements and addressed ethical AI considerations

Freelance ML Engineer & Data Analyst

July 2023 - Feb 2025

- Delivered end-to-end ML solutions and statistical analysis projects for diverse clients using Python, R, and MATLAB across domains including predictive modeling, data pipeline development, and business analytics.

Customer Relations Officer

Dec 2021 - Apr 2022

LB Finance PLC, Delkanda, Sri Lanka

- Managed customer accounts and financial reporting in a fast-paced financial services environment, collaborating with cross-functional teams to resolve customer issues and demonstrating strong stakeholder engagement and problem-solving abilities under pressure

Projects

Multi-Agent Customer Retention System - Azure (Agentic AI & MLOps)

- Built an autonomous AI system using a LangGraph multi-agent architecture and Vector RAG to correlate structured SQL billing data with unstructured complaint logs, calculating dynamic churn risk scores in real-time.
- Deployed a production-ready FastAPI microservice on Azure that enables the agent to autonomously route high-risk cases to human supervisors or instantly negotiate personalized retention offers.
- Tools Used: Python, LangGraph, Google Gemini, Azure Cosmos DB, Azure SQL, Azure Blob Storage, FastAPI, Docker, OpenTelemetry.

Kapruka Malli - E-commerce Shopping Agent (Generative UI & Agentic AI)

- Engineered a Generative UI e-commerce agent utilizing an LLM decision engine and Named Entity Recognition (NER)

to parse unstructured user intent, autonomously selecting from 7 distinct MCP tools across 3 local languages (English, Singlish, Tanglish) with >90% accuracy.

- Integrated a Model Context Protocol (MCP) layer to query live production databases and render dynamic React interfaces with sub-second latency via Edge caching, alongside an automated retention workflow that curates user feeds based on extracted lifecycle events.
- Tools Used: React, Next.js 14, TypeScript, Zustand, Tailwind CSS, Framer Motion, Llama 3, MCP

PaymentGuard: Real-Time Fraud Detection Platform (Microsoft Fabric & Data Engineering)

- Architected a multi-cloud fraud detection data mesh on Microsoft Fabric, virtualizing 285K AWS S3 transactions via zero-copy OneLake shortcuts and processing live telemetry through Eventstream and Eventhouse (KQL) for sub-second anomaly detection.
- Fused Lakehouse Delta tables and real-time KQL streams into Synapse Data Warehouse using T-SQL stored procedures, deploying a Direct Lake Power BI semantic model alongside automated Data Activator incident escalation.
- Tools Used: Microsoft Fabric (Lakehouse, Eventhouse, Synapse Warehouse, Data Activator), PySpark, KQL, T-SQL, Direct Lake Power BI, AWS S3, Python

E-Commerce Customer Intelligence Platform - Azure (MLOps & Business Intelligence)

- Architected serverless Medallion Architecture (Bronze → Silver → Gold) on Azure Databricks processing 100k+ e-commerce orders, achieving cost optimization through PySpark optimization and Delta Lake ACID transactions
- Delivered demand forecasting model (R^2 0.36) and ALS-based recommendation engine after strategic pivot from churn prediction, deploying MLflow-tracked models with Docker and creating Power BI "Whale Detector" dashboard identifying high-value at-risk customers
- Tools Used: Azure (Databricks, Data Factory), PySpark, Delta Lake, Spark MLlib, XGBoost, MLflow, Power BI, Docker

Hybrid Credit Risk Engine & Financial Explainability (Risk & Operations)

- Engineered credit scoring system processing 2.2M+ loan records with Polars streaming, reducing memory overhead by 87% (8GB to <1GB) and achieving 71% AUC with 30.5 KS statistic
- Calibrated LightGBM model to FICO-style scores (300-850) for regulatory compliance, integrating Llama 3.2 for natural language explanations of loan decisions
- Tools Used: Python, LightGBM, Polars, Llama 3.2, Streamlit, Scorecardpy

Achievements

Microsoft Azure Data Scientist & Fabric Analytics Engineer Associate Certifications 2026

- Earned Microsoft Certified: Azure Data Scientist Associate (DP-100) and Fabric Analytics Engineer Associate (DP-600), demonstrating hands-on proficiency in Azure Machine Learning, MLOps, and Fabric data engineering, warehousing, lakehouses, and semantic modeling on Azure.

1st Place - DataXplore 2025 2025

- Won 1st place in a national-level Data Science competition organized by the Statistics Society of the University of Sri Jayewardenepura, competing across multiple domains including Exploratory Data Analysis, Machine Learning, Model Explainability, Time Series Analysis, and Statistical Reasoning.

Research Publication - ICACT International Conference 2025

- Published research on urban computing for sustainable university development at NSBM's 2nd International Conference on Advanced Computing Technologies (ICACT), exploring the application of IoT and machine learning to transportation systems, land use optimization, and air quality monitoring.

Education

BSc (Honours) in Data Science Apr 2022 - Nov 2025

National School of Business Management (NSBM) - Homagama, Sri Lanka

- **First Class Honours** (GPA: 3.76/4.0)
- **Coursework:** Machine Learning, Deep Learning, Big Data Analytics, Statistical Methods, Data Visualization, Cloud Computing, Natural Language Processing

GCE Advanced Level (Physical Science Stream - Mathematics) 2020

Mahinda Rajapaksha College - Homagama, Sri Lanka

Additional Courses

Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300) Certification (In Progress) | MLOps Specialization - Duke University | Advanced Machine Learning on Google Cloud Specialization - Google | Google Advanced Data Analytics Professional Certificate - Google | Meta Data Analyst Professional Certificate - Meta | Databricks Professional ML Practitioner Path | Microsoft Power BI Data Analyst Professional Certificate - Microsoft | Deep Learning Specialization - DeepLearning.AI | Machine Learning Specialization - Stanford University

***References available upon request**